Session 5 Quiz - How AI Learns from Data

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 52625]

The portion of a dataset used to teach the model - what the model adjusts its parameters against - is called the __________ set.

A. Screen

B. Logo

C. Training

D. Cable

Correct Answer: C

Explanation: The training set is the data the model learns from during the training process. Typically 70–80% of total data is used for training, with the rest held back for validation and testing. The other options have nothing to do with ML data splits.

Q2 [Type: MCQ | Difficulty: Easy | QID : 52627]

A separate portion of data, held back from training and used to check how well the model performs on examples it has never seen, is called the __________ set.

A. Test

B. Logo

C. Cable

D. Browser

Correct Answer: A

Explanation: The test set is held back from training and used only after training is complete, to evaluate how the model performs on unseen examples - the closest available proxy for real-world deployment. The other options are unrelated to ML data splits.

Q3 [Type: MCQ | Difficulty: Medium | QID : 52631]

When training data does not include enough examples from all the groups the model will be used on, this is called a __________ problem in the data.

A. Storage

B. Compression

C. Logo

D. Representation

Correct Answer: D

Explanation: Representation problems mean some groups are missing or underrepresented in training data. This typically leads to worse performance for those groups at deployment - one of the most documented sources of AI bias. Storage and compression describe how data is held; they are not the issue here.

Q4 [Type: MCQ | Difficulty: Easy | QID : 52633]

Using the same photos for both training and testing is an effective way to see if an AI truly understands a concept.

A. True

B. False

Correct Answer: B (False)

Explanation: Testing must be done on new data to ensure the AI has learned general patterns rather than just memorising specific examples. A model can score 100% on training data simply by recalling those exact examples, while failing on anything new - which is exactly what the train/test split is designed to detect.

Q5 [Type: MCQ | Difficulty: Medium | QID : 52634]

An AI that has truly "learned" a concept should be able to perform well on a testing dataset it has never seen before.

A. True

B. False

Correct Answer: A (True)

Explanation: The ability to generalise patterns to new, unseen data is the definition of successful machine learning. If a model only performs well on its training data, it has memorised the examples rather than learned the underlying patterns.

Q6 [Type: MCQ | Difficulty: Medium | QID :52635]

Bias in an AI system only happens when the team building it deliberately chooses to discriminate against certain groups.

A. True

B. False

Correct Answer: B (False)

Explanation: Intent is not required for bias. Most documented AI bias was unintentional - produced by data and design choices, not by malice. Early facial recognition systems failed on darker skin not because anyone chose to discriminate, but because the training data was overwhelmingly light-skinned. Amazon's hiring AI replicated past gender bias not because it was told to, but because it learned from historical hiring records that already favoured men.

Q7 [Type: MCQ | Difficulty: Easy | QID :52636]

What term refers to the specific clues or characteristics an AI system uses to find patterns in a dataset?

A. Labels

B. Predictions

C. Features

D. Generalisations

Correct Answer: C

Explanation: Features are the individual characteristics - like colour, shape, or size - that serve as variables for the AI to analyse. Labels are the correct answers attached to training examples. Predictions are the model's outputs. Generalisations describe a model's ability to handle new data, not the inputs themselves.

Q8 [Type: MCQ | Difficulty: Easy | QID : 52637]

Which of these is a "helpful clue" (feature) for an AI trying to recognise a face?

A. The brand of camera used

B. The time of day the photo was taken

C. The colour of the person's shirt

D. The distance between the eyes

Correct Answer: D

Explanation: Distance between the eyes is a consistent anatomical feature that helps distinguish one person from another regardless of clothing, lighting, or location. The other options describe details about the environment or what the person happens to be wearing - useful clues for an overfit model, useless for a model that should generalise.

Q9 [Type: MCQ | Difficulty: Medium | QID : 52638]

Which problem occurs when an AI learns the random details of training examples so well that it fails to work on new data?

A. Overfitting

B. Data bias

C. Underfitting

D. Generalisation

Correct Answer: A

Explanation: Overfitting happens when the AI treats noise or specific background details as important rules, making it inflexible. Data bias is a different problem - unfair representation across groups. Underfitting means the model has not learned enough. Generalisation is the opposite of overfitting - the goal, not the problem.

Q10 [Type: MCQ | Difficulty: Medium | QID : 52639]

"Garbage In = Garbage Out" is a saying that means:

A. The AI will automatically throw out any data that is not useful for learning

B. The AI cleans and organises data before learning patterns from it

C. If the training data is poor quality, the AI's output will also be poor

D. Bad algorithms can always be saved by feeding them good quality data

Correct Answer: C

Explanation: This is the session's cooking analogy - bad ingredients cannot be saved by a good recipe. The quality of an AI's output is directly dependent on the quality of the data used to train it. Options A and B credit the AI with cleanup it does not actually do; option D inverts the relationship.

Q11 [Type: MCQ | Difficulty: Medium | QID : 52640]

Which of the following is NOT one of the "Four Things Good Data Needs"?

A. Complex

B. Clean

C. Varied

D. Fair

Correct Answer: A

Explanation: Data does not need to be costly to be high quality; it needs to be enough, clean, varied, and fair. Cost may sometimes be a side effect of collecting good data, but it is not itself a quality criterion.

Q12 [Type: MCQ | Difficulty: Medium | QID :52645]

What is the likely result of providing an AI with "Limited Data" (only one or two examples)?

A. The AI will generalise perfectly to all new examples it encounters

B. The AI will become "biased" against certain groups of users or inputs

C. The AI will automatically find more data on the internet to improve itself

D. The AI will memorise those specific things and fail on anything else

Correct Answer: D

Explanation: With very little data, the AI has no choice but to memorise the few instances it sees, leading to severe overfitting. It cannot generalise (A) from too little. Bias (B) requires representation gaps across groups, not just data scarcity. AI cannot fetch its own data (C).

Q13 [Type: MCQ | Difficulty: Medium | QID : 52649]

Why does the session describe features as "a detective's clues"?

A. Detectives rely on modern AI tools to solve cases, just as features rely on algorithms to function

B. Features are the pieces of evidence a model uses to make a decision, just as clues guide a detective's conclusion

C. Both detectives and AI systems work best in pairs, just as features work best when combined together

D. Features deliberately hide information from the AI model, just as clues are hidden from detectives at a crime scene

Correct Answer: B

Explanation: The analogy works because features are inputs the model reasons over - like clues a detective examines to reach a conclusion. The other options are decorative (A, C) or contradict what features actually do (D).

Q14 [Type: MCQ | Difficulty: Medium | QID : 52657]

A team has 2 million correctly-labelled photos of skin lesions. The model performs well in trials on patients with light skin tones but poorly on darker skin tones. Of the four pillars of good data - enough, clean, varied, fair - which is MOST clearly missing?

A. Enough - they need to collect several million more labelled photos before training

B. Clean - the labels on the existing photos are likely wrong or inconsistent

C. Varied / Fair - the dataset does not represent all skin tones equitably

D. None - the model just needs more training cycles to improve its accuracy

Correct Answer: C

Explanation: The case stipulates 2 million correctly-labelled examples - there is plenty of data, and it is clean. What is missing is variety across skin tones, which makes the dataset unfair to certain patient groups. A is the lazy default. B contradicts the setup. D doubles down on training the same flawed data.

Q15 [Type: MCQ | Difficulty: Hard | QID :52659]

A team is building a model to predict whether a student will succeed in a coding course. Which feature most clearly risks introducing unfair bias into the predictions?

A. Number of hours spent on previous coding assignments

B. Score on a prior introductory programming quiz

C. Whether the student has completed prior math coursework

D. The neighbourhood or ZIP code where the student lives

Correct Answer: D

Explanation: ZIP code looks neutral but acts as a proxy for income, race, and school quality - the kind of feature that smuggles real-world inequality into model predictions. The other options directly measure relevant academic effort and preparation; ZIP code does not.

Q16 [Type: MCQ | Difficulty: Hard | QID : 52665]

Amazon's hiring AI was trained on 10 years of past hiring decisions and ended up preferring male candidates. Which statement BEST captures why this happened?

A. The algorithm was poorly designed and gave random, unreliable results during candidate evaluation

B. The AI learned the patterns in its training data - including past human biases - and reproduced them

C. The AI "decided" on its own to be sexist against female candidates without any human influence

D. The AI was overfit to the testing set and failed to generalise to real-world hiring decisions

Correct Answer: B

Explanation: Bad data does not just create errors - it encodes existing unfairness. The historical hiring data already favoured men, so the AI learned that pattern. Option A blames the algorithm rather than the data. Option C wrongly attributes intent to a system that has none. Option D does not match the situation, which is about bias, not a train/test gap.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 52667]

Scenario: A model trained on photos from a professional photography studio (consistent backdrop, controlled lighting) achieves 99% accuracy on test photos from the same studio but only 55% on phone snapshots taken in real homes.

Question: Which statement BEST captures what went wrong?

A. The model needs a faster processor to handle real-time predictions in different environments

B. The model's testing split was too large, leaving too little data to learn proper patterns

C. The model used supervised learning when it should have used reinforcement learning for this task

D. The model learned the studio's environment and failed to generalise to other settings

Correct Answer: D

Explanation: This is subtle overfitting: the model could not separate the subject from the studio. The 99% / 55% performance gap between training and real-world data is the giveaway. The other options blame components that have nothing to do with this gap.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 52671]

Scenario: A research team training a heart disease prediction model has 100,000 patient records. They notice 95% of records come from patients over age 60.

Question: What is one important implication?

A. The model will work equally well on patients of all ages

B. Age does not affect model performance, so the imbalance is not a problem

C. The model may perform less reliably on younger patients

D. The team should remove age from the model entirely

Correct Answer: C

Explanation: Heart disease presentation differs by age, and the model has very few young-patient examples to learn from - a classic representation gap. Skewed training data produces skewed performance. Option A ignores the imbalance. Option B is factually wrong - patient demographics affect models trained on patient data. Option D does not address the underlying issue.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 54519]

Scenario: A small clinic is excited to use a free AI tool that helps diagnose common conditions. They notice the tool was developed using data from large urban hospitals. Their clinic is in a rural area with a different patient population.

Question: Explain one concern they should consider before relying on the tool.

(Respond in 3–4 sentences.)

Model Answer: The tool was trained on patterns from urban hospitals, but rural patients may differ in important ways - different demographics, different prevalence of certain conditions, different access patterns, and different symptom presentations based on access to earlier care. The model may perform less reliably on the clinic's patients because they are different from the training population. The clinic should validate the tool's accuracy on their actual patients before relying on its outputs for clinical decisions.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 54520]

Scenario: A national park service is considering training an AI to identify endangered species from camera-trap footage across multiple parks. The AI would help rangers prioritise where to focus protection efforts. The training data would come from existing camera traps. Some scientists are concerned that the training data over-represents species in well-studied parks and underrepresents species in less-studied parks where endangered species often live.

Question: Identify one significant benefit AND one significant risk, and describe what the park service should require before deployment.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that AI image identification can process far more footage than rangers could manually, helping prioritize protection efforts in remote or hard-to-monitor areas and potentially identifying species at risk that would otherwise go undetected. A significant risk is that an AI trained mostly on well-studied parks will perform best at identifying well-studied species - meaning the system could systematically miss endangered species that are precisely the ones most needing protection, while confidently identifying common species and creating a false sense that fewer rare species exist when the AI just cannot see them. The park service should require representative training data from less-studied parks including known images of the actual endangered species, validation with field experts on each park's species, transparency about which species the system can and cannot reliably detect, regular auditing as new data accumulates, and human ranger review of all conservation decisions - recognising that for endangered species conservation, missing a real sighting can have severe consequences and the system should not be relied upon for "no detection" conclusions without human verification.